THE USE OF IDEF0 FOR THE DESIGN AND SPECIFICATION OF METHODOLOGIES

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ABSTRACT

This paper presents experiences with the application of the IDEF0 technique in the development and specification of methodologies. Case studies in this novel use of the tool are presented for two projects. The first case describes a methodology for assisting small companies implement continuous improvement. The second case is a methodology for the strategic justification of investments in enterprise-wide technologies. The conversion of the IDEF0 specifications to actual delivery mechanisms is discussed. Advantages, disadvantages, and suggestions for use of IDEF0 for methodology design and specification are also presented.

INTRODUCTION

IDEF0 is a widely used technique for the structured analysis and design of systems. Its use in improving the productivity and communications in computer integrated manufacturing systems and, more recently, as a tool for business process reengineering efforts are widely documented. In this paper, we present what we believe is a novel application for IDEF0, the design and specification of methodologies. A methodology in the context of this paper can be thought of as the combination of actions, methods, and tools used together to achieve some purpose. Examples include methodologies to develop manufacturing systems, implement organizational change, or manage projects. All of these include some prescriptions of the steps required, and the methods and tools (computerized and otherwise) supporting the steps Take as an example a methodology for business process reengineering. Such a methodology would include the steps required (set process goals, document current processes, etc.) along with the tools and methods (process modeling, modeling software, group decision making techniques, etc.) to be utilized at various steps.

The primary advantage of a well documented methodology is in the structure it provides. The presence of well documented methodologies allow novices to accomplish a given task with greater efficiency and effectiveness. The structure provided by a formally specified methodology also leads to repeatability of success and better integration of the work of multiple developers.

Current efforts in the development and specification of methodologies appear to be largely an ad hoc process. In this paper we describe our experiences in developing two methodologies. The first, entitled "Perform Continuous Enterprise Improvement" is a methodology designed be used by small manufacturing enterprises in implementing continuous improvement principles. The second is a methodology for the strategic justification of integrated enterprise systems. Both of these were developed and documented with the aid of IDEF0, a structured modeling and analysis technique. We believe that the use of a tool such as IDEF0 greatly enhances the success of a methodology development project.

This paper begins with a history and overview of the IDEF0 technique. It then presents an overview of

the two methodologies and their development. The paper concludes with a discussion of the strengths and weaknesses of using IDEF0 in this environment.

HISTORY AND BACKGROUND OF IDEF0

IDEF0 has its beginning with Structured Analysis and Design (SADT). Softech, Inc. developed SADT in the mid-1970s in an effort to overcome some of the shortcomings of the modeling and analysis methods of that time [8]. In the late 1970s, the Air Force selected SADT as the language to support its Integrated Computer Aided Manufacturing (ICAM) initiative [4]. It was from this effort that IDEF0 was developed and brought into wide use, especially within the aerospace industry. Since that time much has been published in literature concerning the uses of IDEF0 and SADT. According to Ross [7], the methodology has been used by thousands of people from hundreds of organizations. Project areas have included system definition and design, project management and integration. Colquhoun et al. [2] present an excellent review of applications to which IDEF0 have been applied. An organization for practitioners and researchers in IDEF methods, the IDEF Users Group, has been formed. This group holds regular meetings and conferences at which presentations and papers detailing research and case examples of IDEF uses are presented. The number of participants in these meetings and the number and quality of papers presented at the conferences are anecdotal evidence of the widespread adoption of the IDEF method.

IDEF0 is a functional enterprise modeling tool that was developed through the Air Force's Integrated Computer Aided Manufacturing (ICAM) program. IDEF stands for ICAM DEFinition. IDEF0 is used to represent the *functional* (i.e., activity or process oriented) framework of a system.

There are five elements to the IDEF0 functional model (see Figure 1): the activity (or process) is represented by boxes; inputs are represented by the arrows flowing into the left hand side of an activity box; outputs are represented by arrows flowing out the right hand side of an activity box; the arrows flowing into the top portion of the box represent constraints or controls on the activities; and the final element represented by arrows flowing into the bottom of the activity box are the mechanisms that carries out the activity. The inputs, control, output and mechanism arrows are also defined as ICOM's [3, 4].

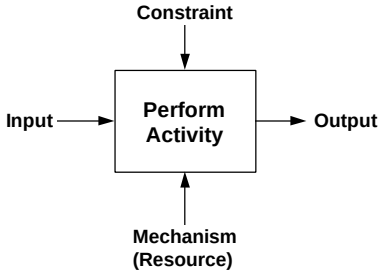

Figure 1. IDEF0 Representation

Another characteristic of the $IDEF_0$ modeling technique is that each activity and the ICOM's can be decomposed (or exploded) into more detailed levels of analysis. This is seen in Figure 2 below. These characteristics will be evident in the graphical description of the two methodologies described further in this paper.

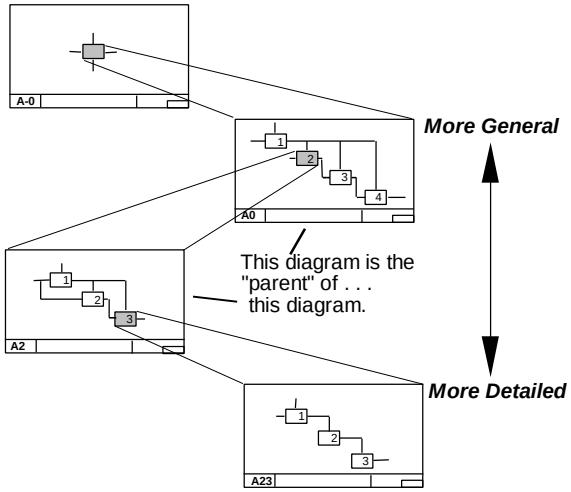

Figure 2. Decomposition Overview

CASE STUDIES

This section will detail two case studies of using the IDEF0 method to develop and document methodologies. The two methodologies are: 1) Perform Continuous Enterprise Improvement (PCEI), a methodology to aid in the implementation of continuous improvement at small companies; and 2) Perform Strategic Justification of Integrated Enterprise Technologies (SJET), a methodology for the strategic justification of integrated enterprise systems. A brief

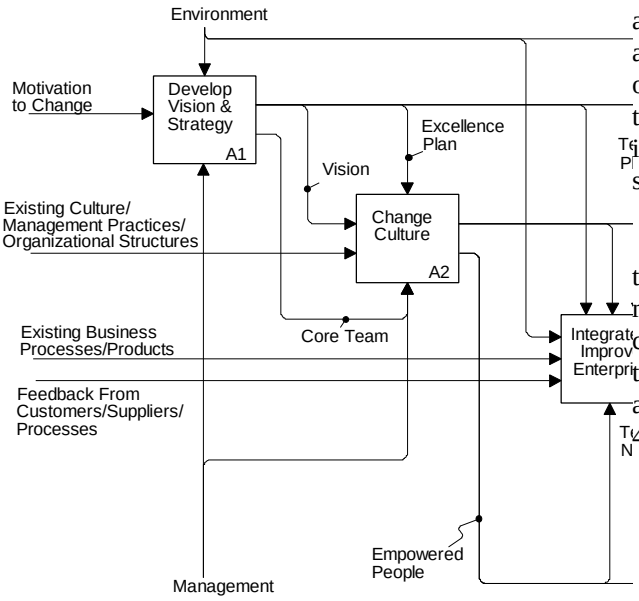

Figure 3. Perform Continuous Enterprise Improvement

description of each methodology will be provided. The bulk of the discussion, however, will be on the development efforts and the applicability of IDEF0 to the process. Experiences with the use of IDEF0 and how the methodologies developed were converted to deployment tools are presented. Advantages and disadvantages with the use of the technique are presented.

Perform Continuous Enterprise Improvement

The focus of this methodology is on the implementation of the concepts of continuous improvement at small- to medium-sized manufacturing enterprises. The model [1, 6] describes the four basic elements to a continuous improvement program: the development of a coherent vision and strategy, cultural change to support the vision, the continuous improvement and functional integration of the enterprise, and the development of technology solutions (Figure 3). It is important to point out that although the activities are shown to be chronological, there is constant feedback among the activities. Enterprise excellence is, as the name continuous improvement implies, a never-ending process.

The model was developed based on a review of literature and the experiences of a team of approximately 10 researchers. The top level of the diagram (Figure 3) was first developed by the entire team. The model was then divided, with a team assigned to decompose each of the boxes. The first activity was decomposed to one further level, the others were broken down into two further levels. The top level diagram was modified several times to take into account additional knowledge gained during the subsequent work.

The actual delivery of the methodology is not through the IDEF0 model. It was believed that the model provided more detail than necessary to the clients receiving assistance. Also, the complexity of the model was deemed to be a deterrent to acceptance. A stylized version of the model (Figure 4) is used in discussions with clients.

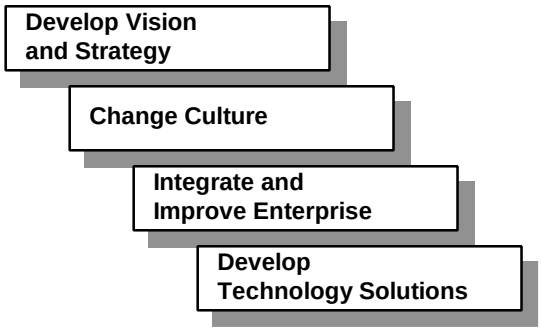

Figure 4. Enterprise Excellence Methodology

The actual deployment is accomplished through a program entitled "The Journey to Excellence". This program incorporates the concepts and practices detailed in the model into a series of workbooks and presentations. Initial versions of the program consisted of "Phases" roughly corresponding to the four boxes or activities at the top level of the model. The iterative and concurrent nature of the continuous improvement process, however, led to a change in the program where the boundaries of the four activities are less distinct. The development of the workbook and presentations from the IDEF0 specification in this case was rather loose. The specification provided the activities which had to be performed. The actual mechanics for the activity are then developed in the workbook and programs. For example, the initial activity in the model is Develop Vision and Strategy. In implementing the methodology at client companies, this activity is accomplished through a series of "Executive Planning Sessions" in which management and workers of the firm are led through a workshop consisting of a fairly typical Strengths -Weaknesses - Opportunities - Threats (SWOT) analysis with special emphasis on continuous improvement and customer focus.

The methodology has been applied with success at over 20 small manufacturers. Results realized have included improved communications and morale, reduced work in process, reduced scrap, improved throughput, and successful implementation of cellular manufacturing technologies.

Strategic Justification of Enterprise Technologies

The second methodology to be discussed is entitled Strategic Justification of Enterprise Technologies [5]. This methodology was developed under funding from the National Center for Manufacturing Sciences in response to a perceived need to have a comprehensive methodology for pervasive and enterprise-wide technologies which possess strategic benefits. The adoption of integrated enterprise technologies is a key to continued competitiveness in the world market. However, justification of these technologies is difficult when they are judged solely on traditional economic criteria. The failure to consider their strategic and long-term benefits has led to the failure of manufacturers to adopt these technologies. Another complicating factor in the justification and adoption of integrated technologies is the cultural and organizational issues involved. The impact of implementing, for example, an MRP II system crosses many organizational boundaries. The success or failure of this implementation depends on the buy-in of all organizations and individuals involved. Traditional methods of justification often do not consider these organizational impacts and are not designed for group consensus building.

The development of this methodology was accomplished through a series of meetings with "Customer Teams" - individuals who were potential users of the final methodology. Quality Function Deployment (QFD) was used to determine the customer team's requirements for the methodology and supporting deployment tools. The QFD analysis was used in then developing and documenting the methodology using IDEF0 [9].

An initial high-level decomposition was first created and approved. At the highest (or A0) level, five activities were identified (Figure 5). These "Ax" activities are: Identify System Impact, Identify Transition Impact, Estimate Costs and Benefits, Perform Decision Analysis, and Audit Decision. Each of these Ax activities were then decomposed into two further levels (Axx and Axxx). Each of the Ax activities were decomposed to its lowest level before beginning another activity. The activities were analyzed in the order the developers felt would best facilitate the development of the methodology rather than in consecutive order. The actual sequence of decomposition was: Identify System Impact, Perform Decision Analysis, Estimate Costs and Benefits, Identify Transition Impact, and Audit Decision. A total of 62 activities were eventually described.

The development of the methodology was accomplished through an "author-reader cycle" in which the developers incrementally created "kits" consisting of a diagram and accompanying text for the activities required for the justification process. These were then circulated to members of development and customer teams for comment. The comments were then incorporated into the final methodology documentation. The model was also reviewed by the customer teams at several meetings. Comments from these meetings were also incorporated [9].

As with the PCEI model, the IDEF0 model was not the delivery mechanism but rather its design specification. The delivery of the methodology is accomplished through a workbook and supporting software decision support system. The conversion to the workbook was much more formal in this case than with the PCEI model. As the complete decomposition of each Ax activity to all its Axxx activities was completed and approved, the text was transferred to the workbook. The transfer from IDEF0 to the workbook required minimal effort. Each Ax activity became a "Phase" or chapter in the workbook, each Axx activity a "Task", and each Axxx activity a "Step". Much of the text in the IDEF0 model was design oriented and not appropriate to the target audience of the workbook. Therefore, the text was made less formal and practitioner oriented. Care was taken to ensure that the intent of the IDEF0 model text was not lost. Several reviews were performed to validate the consistency of the workbook and model. Additionally, material not included in the IDEF0 text but necessary for the workbook was also created. Actual forms and templates that an analyst would use in completing a justification analysis were created. A step-by-step, running example was created to illustrate and clarify the methodology. In some cases, the examples from the model were expanded. Suggested readings for each Phase were also identified.

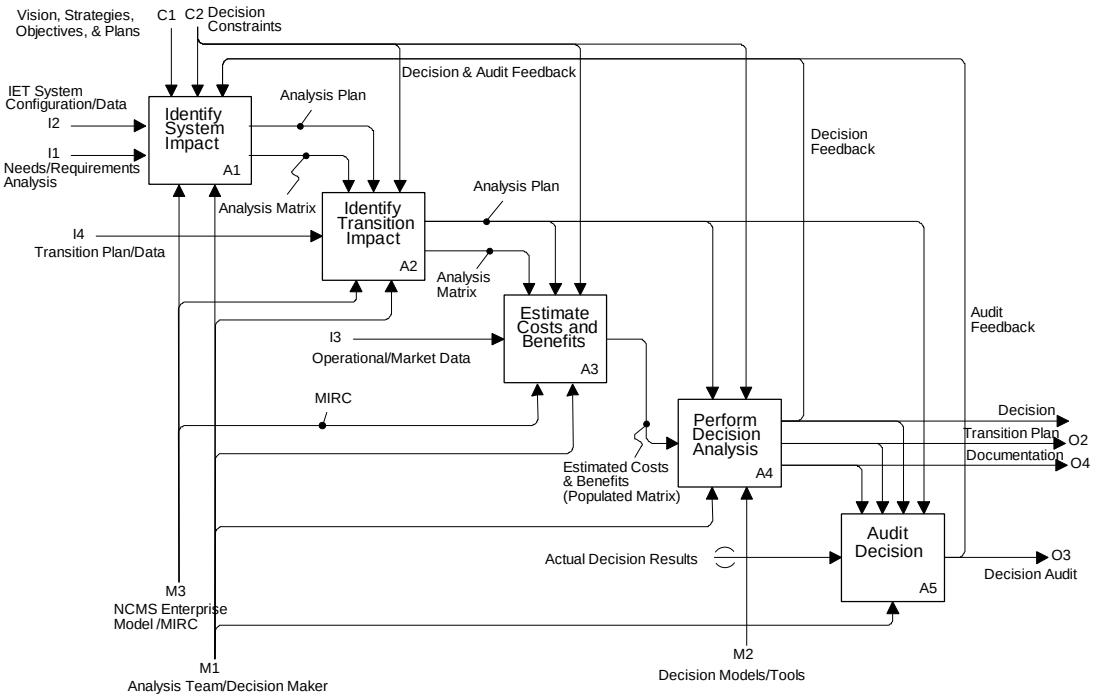

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Figure 5. Perform Strategic Justification of Enterprise Technologies

The methodology is realized as a set of forms and matrices which lead the user through the justification process. Relevant information to the justification are first documented in forms. Based on the information in these forms, matrices are created which link the technology under consideration to the enterprise, both in terms of activities affected and strategic areas impacted. The methodology allows for the integration of strategic metrics with more traditional cost metrics in arriving at a final "score" for the alternative. The methodology encourages and supports a group-based approach to justification. Included in the methodology are explicit considerations of importance rankings and utility functions to allow for the capture of the decision maker's (either an individual or group of individuals) preferences.

The development of the computer system was tied to the IDEF0 model mostly through the workbook. This system basically automates the manipulation of the forms and matrices. The system is meant to augment the workbook, not replace it.

The methodology, workbook, and software are currently undergoing field tests at several locations. They are being applied to actual justification problems at these companies. Feedback to date has been generally favorable. Comments and issues arising from the field tests will be incorporated into a final revision.

In addition to the two projects discussed, we have used IDEF0 for several other non-manufacturing system description efforts. It has been used extensively in project management in which the planning and coordination of tasks are aided by the decomposition facility of IDEF0. For example, each model level can be made to correspond to a level of a work breakdown structure. It has also been used for specifying a model describing the management of training program at a small company and is currently being used to develop a methodology for the development of strategic process performance measures, in which a large number of consultants are participating.

ADVANTAGES AND DISADVANTAGES OF USING IDEF0

The major advantage with using IDEF0 is the formalism that it imposes on the user. This formalism leads to the creation of consistent, integratable models. In our experiences, many different researchers were involved in development efforts. Using IDEF0, the parts of the models produced by each researcher were easily integratable into the larger model. By partitioning the model into parts and defining the interfaces involved, each part of the model can be developed independently.

The IDEF0 development methodology consists of progressively decomposing a system from a high level description. This process was found to work very well for methodology development. Each model was begun with a concept of what had to be done. The step-by-step development (with feedback and changes) helped insure the inclusion of all necessary activities in the two methodologies.

The use of review kits was also found to be very advantageous. Kits allow for the inclusion of comments from a large number of experts and developers. The formal process of kits consists of reviewers making comments to the kits and the developers either incorporating the comments or responding to the reviewer on why the comment was not incorporated. The main contact between developer and reviewer is through the kit. This allows for a more efficient process by forcing both the reviewer and developer to be concise. Again, this allows for the integration of comments from a large number of reviewers. Issues related to reaching consensus in a large group can be largely avoided.

Another advantage in using a tool such as IDEF0 is its wide acceptance. With a few small exceptions, the models developed conformed to IDEF0 standards. The familiarity of many users with the syntax meant less explanation and training of reviewers was required.

Several automated tools are available to support the development of IDEF0 models. These tools include those which run on popular platforms including both Macintosh and DOS based personal computers. This support proved invaluable in the development and conversion of the methodologies discussed. The text and diagrams developed could be easily copied into word processing and other applications. Likewise, text and diagrams from other applications could be inserted into the IDEF0 model. In the case of the SJET application, much of the text which made its way into the workbook was copied directly from the model. The matrices were developed in a spreadsheet program and used directly in the model and workbook.

With IDEF0 models, the placement of boxes does not imply a strict precedence sequence. This, and the ability to specify explicit feedback between activities was found to be usefulness in the development of our methodologies in which several activities can occur concurrently and the actual sequence in which tasks are accomplished may be variable depending on a specific implementation.

There were some disadvantages and issues encountered in the use of IDEF0 for this purpose. One of the disadvantages is the constraints put on the user by the technique. While the formalism and rigor proved to be a major assistance, the danger of becoming too caught up in the technique was encountered. For example, one of the rules in IDEF0 is that each diagram should consist of between 3 to 6 boxes. In the SJET methodologies, instances were encountered where it was difficult to describe more than 2 activities without becoming trivial. However, because of the workbook structure being developed, it still made sense to keep the part of the model as a separate activity. Because of this the 2 box rule was violated in the model on two instances. We do not believe this compromised the model in any way.

Another disadvantage is the aversion that some reviewers and end users express when first presented with an IDEF0 diagram. The network of boxes and arrows, along with the size of some models, can cause many users to reject the model. In our experience, most will overcome this initial reaction if the modeling syntax is explained to them. However, because of the large numbers of people we foresaw would be seeing the models, stylized versions of each were produced. Also, it was not necessary to show all users the detail required in the model specifications.

The complexity of IDEF0 can lead to a learning period for new modelers. This learning period is seen in first understanding IDEF0 itself and in then learning to use the computer applications supporting it. In our experience, some formal training is required in the technique itself. Most of the computer systems are simple enough to use with some hands on experience.

RECOMMENDATIONS AND FINDINGS

In our experience, the use of IDEF0 in developing and documenting methodologies offers advantages which outweigh its disadvantages. Instances in which it is especially helpful include complex methodologies which can be decomposed into smaller tasks. It is especially useful in cases in which several different developers are assigned to work on certain parts of the model and where the input of outside experts are required, especially if they are geographically dispersed and circumstances make it difficult to bring them together at one time. But again, it is not without disadvantages. We have not used IDEF0 for every project encountered. The merits of using IDEF0 should be considered on a case by case basis.

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REFERENCES

1. Automation & Robotics Research Institute. *Perform Continuous Enterprise Improvement*. Automation & Robotics Research Institute, Fort Worth, TX, (1991).

2. Colquhoun, G.J., R.W. Baines, and R. Crossley. "A State of the Art Review of IDEF0". *International Journal of Computer Integrated Manufacturing*. 6, 252-264 (1993).

3. Mayer, R., *IDEF0 Functional Modeling*. Knowledge Based Systems, Inc., College Station, TX, (1990).

4. Mayer, R.J., *IDEF0 Function Modeling - A Reconstruction of the Original Air Force Wright Aeronautical Laboratory Technical Report - AFWAL-TR-81-4023*. 1st ed. Knowledge-Based Systems, Inc., College Station, Texas, (1992).

5. National Center for Manufacturing Sciences (NCMS). *Perform Strategic Justification of Integrated Enterprise Technologies*. NCMS, Ann Arbor, MI, (1994).

6. Presley, A., M. Johnson, J. Weddle, and D. Liles. "Enterprise Excellence: Small Manufacturers and Continuous Improvement" in *The 2nd Industrial Engineering Research Conference*. Los Angeles, CA (1993).

7. Ross, D.T. "Applications and Extensions of SADT". *Computer*. April, 25-34 (1985).

8. Ross, D.T. and K.E. Schoman. "Structured Analysis for Requirements Definition". *IEEE Transactions on Software Engineering*. SE-3, 6-15 (1977).

9. Sarkis, J., M. Johnson, A. Presley and D. Liles. "Development of the Requirements for a Strategic Justification Methodology". *Intelligent Automation and Soft Computing, Proceedings of the World Automation Congress*. Albuquerque, NM, TSI Press. 87-91 (1994).

BIOGRAPHICAL SKETCHES

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